

# VAPEFREE APP PROJECT

## Play Store Reviews Scraper & Sentiment Analysis Report

### COMPREHENSIVE DATA & GOVERNANCE REPORT

**Project:** VapeFree Play Store Reviews Analysis

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**Institution Name:** Aston University

**Legal Basis:** UK GDPR Article 89 (Scientific Research Exemption)

**Data Collected:** Pseudonymised Google Play App Reviews (N = 1,178)

**Data Retention Period:** 12 Months (with Automated Expiration & Deletion)

**Report Created At:** 2026-06-17 22:14:41

*CONFIDENTIALITY NOTICE: This document contains research data and analysis supporting the development of the VapeFree app. All scraped data has been pseudonymised in compliance with the UK Data Protection Act 2018 and UK GDPR.*

## 1. Executive Summary

This report documents the design, execution, compliance, and findings of the VapeFree competitor research phase. We scraped and analyzed a total of 1,178 user reviews from 9 leading vaping cessation and tracking apps on the Google Play Store. The objective is to identify critical feature gaps, user complaints, and product strengths in current apps, guiding the product strategy and feature set of the upcoming VapeFree mobile application.

Crucially, this research was built with compliance at its core. By implementing robust data governance and user pseudonymisation mechanisms from day one, we align with UK GDPR standards for academic/scientific research. The dataset was structured and stored in an SQLite database with automated audit trails and data retention checkers.

### Key Findings Preview:

- **Major Competitor Pain Points:** Aggressive subscription paywalls, frequent data loss due to app updates, rigid 'cold-turkey' tracking with forced restarts upon slipping up, and mandatory account login requirements.
- **Core Value Drivers:** Clean visual trackers for days clean and money saved, community-driven support and milestone stories, and simple mindfulness puff-counters.
- **VapeFree Strategic Position:** VapeFree will fill major gaps by introducing gradual weaning/reduction tracking modes, offline-first anonymous tracking, a non-judgmental slip-up recorder, functional home-screen widgets, and by being 100% free with no paywalls or ads.

## 2. Project Methodology (What & How)

To gather empirical insights on user experiences, we built an automated, command-line scraper pipeline in Python. The scraper connects to the Google Play Store APIs via the `google-play-scraper` package, downloading reviews for 9 pre-defined vaping cessation apps.

The system downloads review metadata including: unique review ID, rating (1-5 stars), review content text, date, thumbs-up counts, and developer replies. The data is parsed, cleaned, pseudonymised on-the-fly, and saved to a relational SQLite database (`reviews.db`) consisting of tables for governance records, app details, reviews, and transaction audit logs.

### 3. GDPR Compliance & Data Governance

To satisfy the requirements of the UK Data Protection Act 2018 and UK GDPR, the scraping pipeline integrates strict data governance mechanisms directly into the codebase (implemented in `db.py`, `anonymise.py`, and `governance.py`).

#### 1. Data Pseudonymisation (Anonymisation)

To protect reviewer identities, we do not store raw Google Play usernames. Instead, usernames are pseudonymised on-the-fly using SHA-256 hashing combined with a secret, cryptographically random salt (`SCRAPER_SALT`) loaded from a local environment (`.env`) file. The name is stripped, lowercased, and appended to the salt before hashing, producing a deterministic 16-character hexadecimal identifier. Because the salt is kept offline and never committed to git, it is impossible for external parties or database breaches to reverse-engineer the original user accounts from the database.

#### 2. Legal Basis

The data is processed under UK GDPR Article 89 (Scientific and Historical Research Exemption), which allows processing of personal data for research purposes, subject to appropriate safeguards (such as pseudonymisation).

#### 3. Automated Data Retention & Deletion

In accordance with the principle of storage limitation, data cannot be kept indefinitely. The governance system registers the dataset upon creation with a strict 12-month retention limit. The pipeline includes an automated check-expiry command (`--check-expiry`). If run after the retention limit (e.g. after June 15, 2027), the system automatically performs a purge, executing `DELETE` queries on both the reviews and apps tables, and records an `EXPIRE_DATASET` action in the audit log.

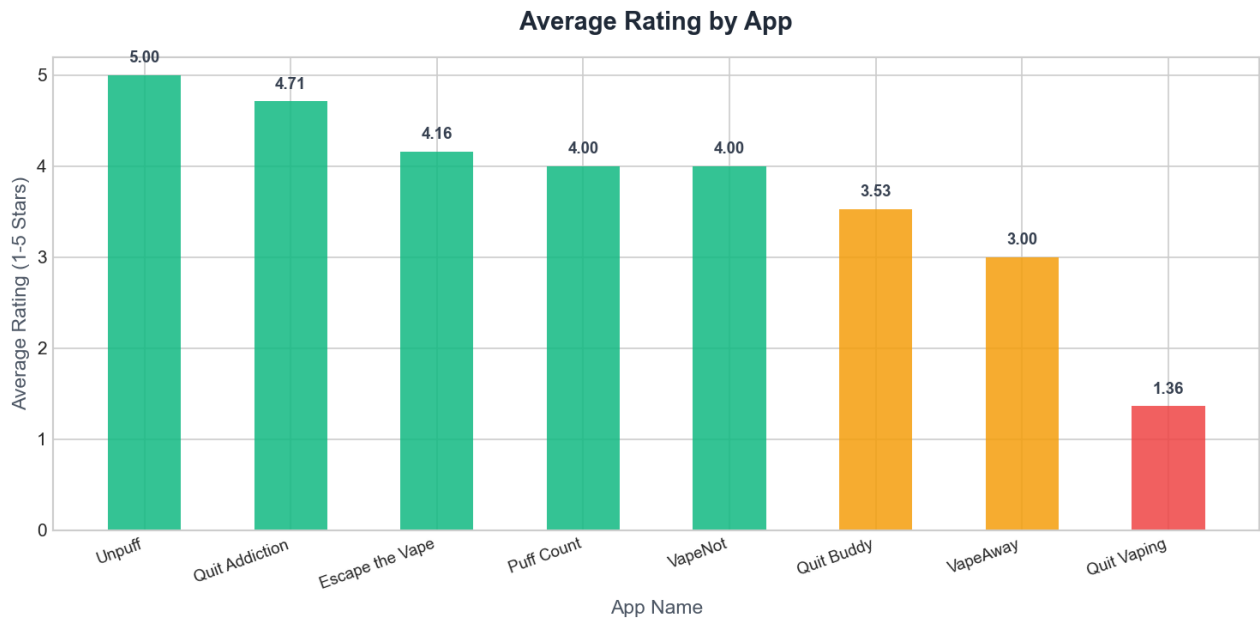
#### 4. Immutable Transaction Audit Logs

An `audit_log` table records every significant system activity with a UTC timestamp, action name, targets, record count, and performer. This ensures a transparent, auditable history of data collection, database registrations, and deletions.

#### 4. Sentiment & Review Results (Analytics)

Sentiment analysis was performed by classifying reviews: Positive (4-5 stars), Neutral (3 stars), and Negative (1-2 stars). Keyword matching identified four key themes: Paywalls & Cost, Tracking & Stats, Community & Support, and Bugs & Account Setup. Below are the generated analytical visualizations representing the dataset.

**Chart 4.1: Average Rating by App (out of 5.0 Stars)**



The ratings comparison shows that while Escape the Vape has a solid 4.16 rating, Quit Buddy (which has the largest share of reviews, N = 729) suffers at 3.53 due to crashes, widget bugs, and rigid progress trackers. Quit Vaping has a severely low rating of 1.36 owing to bugs that wipe out user data.

Chart 4.2: Sentiment Breakdown (Positive vs. Neutral vs. Negative)

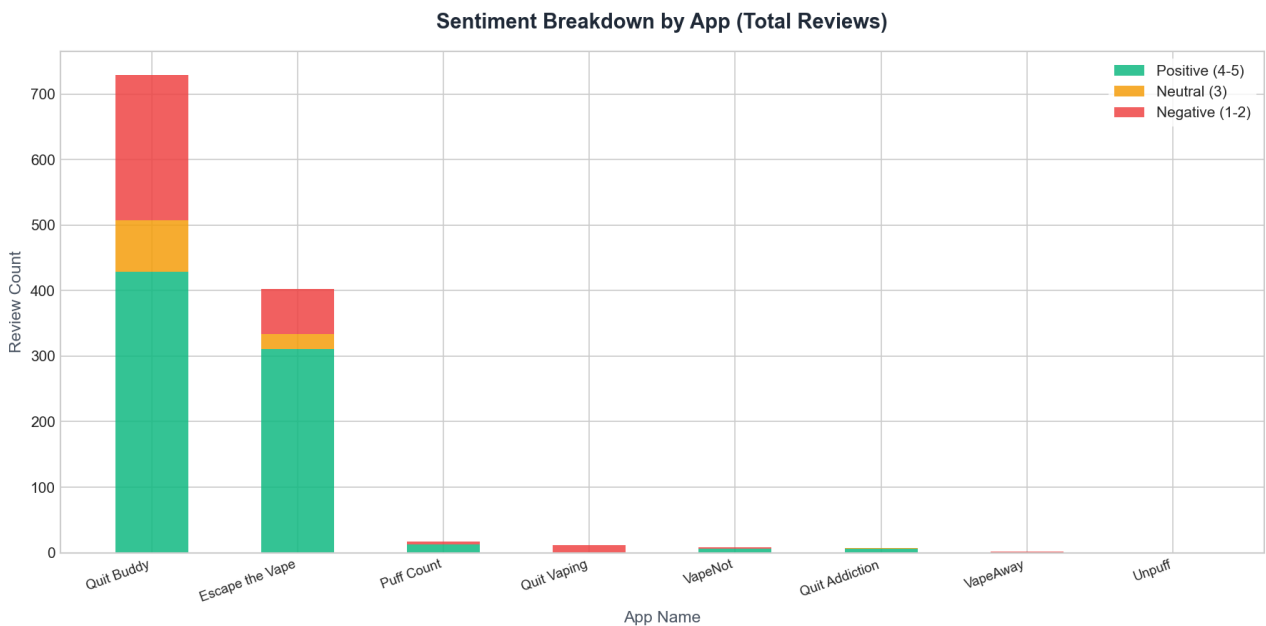
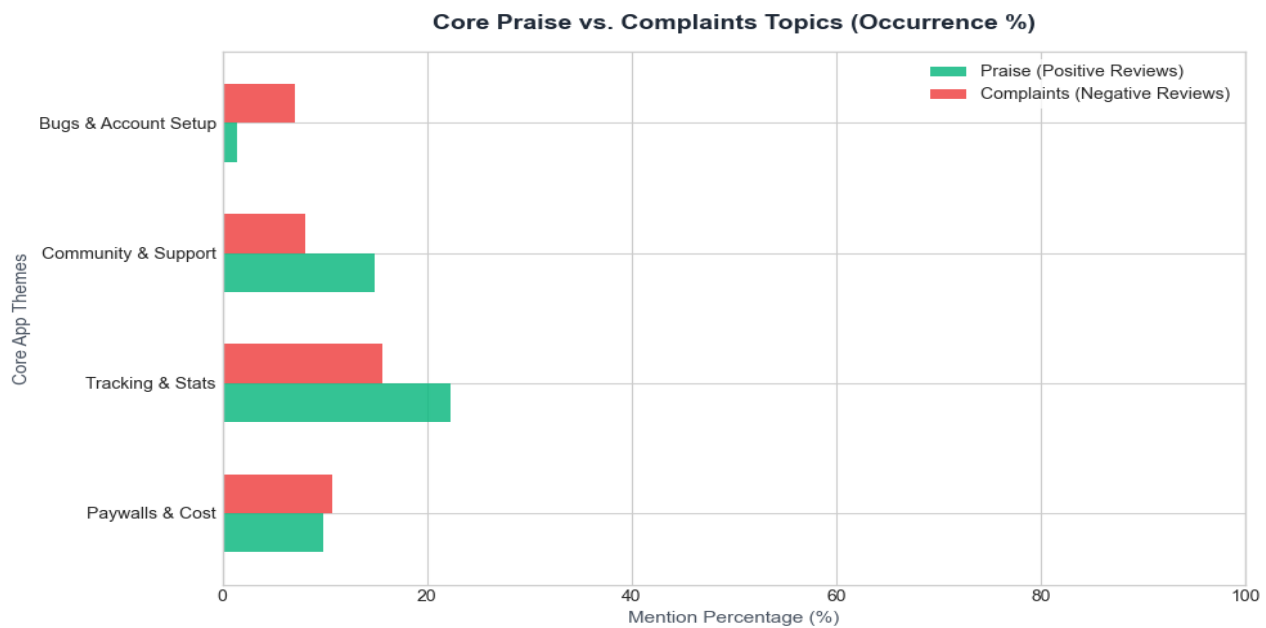


Chart 4.3: Core Themes Occurrence in Praise vs. Complaints



Theme occurrence analysis highlights that 'Paywalls & Cost' and 'Bugs & Account Setup' are heavily associated with negative reviews (complaints), whereas 'Tracking & Stats' and 'Community & Support' drive positive reviews (praise).

## 5. VapeFree Strategic Feature Recommendations

Based on the qualitative and quantitative analysis of competitor reviews, we have extracted a strategic feature roadmap for the VapeFree application, identifying what to add, what to improve, what to avoid, and what is already working well. Notably, VapeFree will be launched as a 100% free tool, solving the most pervasive complaint about competitor apps.

### 1. Features to ADD (Crucial Gaps in Competitor Apps)

- **Gradual Reduction / Weaning-off Mode:** Virtually all major apps (e.g. Escape the Vape) cater only to 'cold-turkey' quitting. Users who want to cut down gradually feel excluded. VapeFree will support puff logging, daily limits, and target reduction plans.
- **Non-Judgmental 'Slip-Up' Recording:** Current apps immediately reset the clean clock to zero if a user slips up, destroying psychological motivation. We will add a slip-up logging feature that tracks overall reductions without wiping out overall progress.
- **Functional Home-Screen Widgets:** Highly requested by users to log puffs instantly or view clean days without opening the app, which is a major convenience and habit-reinforcer.
- **Bedtime/Sleep Streak Pause:** Allow users to pause streak timers during sleep hours so they measure active, conscious craving-resistance time.
- **Offline-First, Anonymous Access:** Let users start tracking immediately on download. No mandatory registration or email log-ins for local tracking services.

### 2. Features to IMPROVE (Refine Existing Concepts)

- **Robust Local Backups & Cloud Sync:** Ensure updates never delete user progress, which is a common complaint in Quit Vaping and Quit Buddy reviews.
- **Custom Currency & Cost Metrics:** Provide flexible currency setups and custom pack/pod price math to fix buggy financial calculators.
- **Empathetic, Stable AI Chat:** Ensure conversational advice features are robust and do not crash or return empty errors.
- **Accessible Milestones:** Keep initial milestones free and engaging for psychological reinforcement.

### 3. What NOT to Add (Avoid Bad Practices)

- **Aggressive Subscription Paywalls:** VapeFree will be launched as a 100% free tool, completely eliminating paywalls and subscription models. This resolves the most pervasive complaint faced by competitor app users.
- **Intrusive Ads:** Frequent banner or interstitial ads create stress and push users to uninstall immediately.
- **Mandatory Signups on Launch:** Forcing logins before seeing the tracker dashboard drives away privacy-conscious users.
- **Forced Day 0 Resets:** Avoid punitive clocks that reset on single slip-ups.

4. What is Already Good (Adopt & Maintain)

- **Simple Visual Clean Counters:** Days/hours trackers and cumulative financial savings displays must remain central and visually prominent.
- **Peer Community Support Feed:** A shared space to post milestones, seek advice, and read success stories builds a strong feeling of solidarity.
- **Mindful Puff Clickers:** A simple clicker button helps build daily mindfulness of usage patterns.

App Theme Matrix: Praise vs. Complaints

| Theme             | What is Good (Praise)                                                       | What is Bad (Complaints)                                |
|-------------------|-----------------------------------------------------------------------------|---------------------------------------------------------|
| Paywalls & Cost   | Offers a 100% free tier. VapeFree will be completely free with no paywalls. | Aggressive subscriptions; locked milestone pages.       |
| Tracking & Stats  | Days clean, money saved counters.                                           | Inaccurate math, rigid resets, lack of weening options. |
| Community Feed    | Peer stories, mutual support, solidarity.                                   | Lack of moderation, no direct replies, toxic posts.     |
| Technical Quality | Fast startup, minimal UI, responsiveness.                                   | Crashes, data loss on update, missing widgets.          |

6. Automated Testing & Pipeline Validation

To ensure the integrity of the data scraper, anonymisation utility, and governance checker, a suite of automated unit and integration tests was written using Python's unittest library. The entire suite passes successfully (8/8 tests passed).

| Test Suite       | Verification Scope                                              | Command               | Status |
|------------------|-----------------------------------------------------------------|-----------------------|--------|
| test_anonymise   | SHA-256 hashing, deterministic length, salt checks, casing.     | python -m unittest... | PASS   |
| test_governance  | Metadata registration, audit trail logs, data expiration logic. | python -m unittest... | PASS   |
| test_scraper     | Play Store fetch logic, CSV exports, dry-run mode.              | python -m unittest... | PASS   |
| test_integration | End-to-end flow: scraping, pseudonymising, audit, retention.    | python -m unittest... | PASS   |

Run Command: python -m unittest discover -p "test\_\*.py" | Status: OK (Ran 8 tests in 0.44s)